■ SOIL LAYERS — MAKE A CORE SAMPLE ■

A Science Lesson for Young Explorers

Detailed Teacher & Student Reference Guide

■ Learning Objectives

- Name and describe the **3 layers of soil** (Topsoil, Subsoil, Parent Rock)
- Understand the role and characteristics of each soil layer
- Build a **model core sample** to represent soil layers
- Identify **living organisms** found in topsoil
- Connect soil science to everyday Indian life and nature

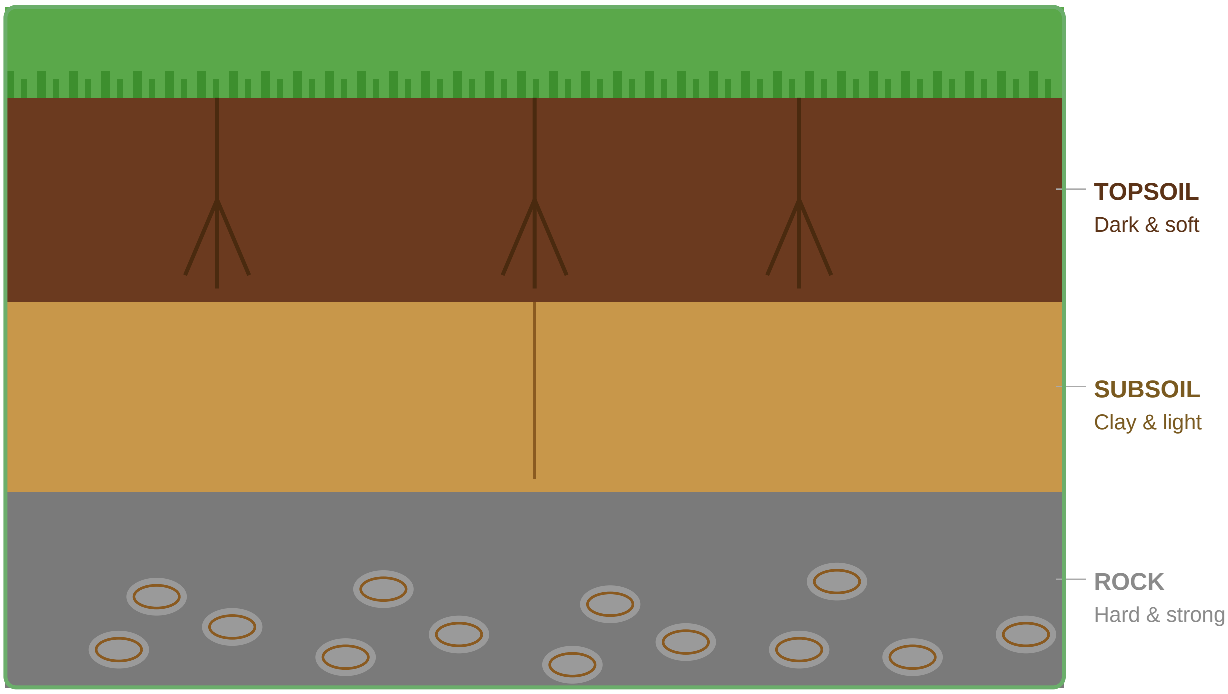
1. Introduction: What Is Under Our Feet?

Every time we walk, play cricket, or tend a garden, we stand on something amazing — **soil**! But soil is not just one thing. Like a delicious **barfi** with layers of different flavours, the ground beneath us has distinct layers, each with its own colour, texture, and purpose.

Scientists who study soil are called **pedologists** or **soil scientists**. They use special tools called **core samplers** to take a vertical slice of the earth and study all the layers at once — no digging required!

■ Did You Know?	It takes **500–1000 years** for nature to build just 2–3 cm of topsoil! Topsoil is one of Earth's most precious resources.

2. The 3 Layers of Soil — Visual Overview



3. Detailed Layer Descriptions

■ Layer 1: TOPSOIL (Top Layer)	
Colour	Dark brown or black — like rich garden soil
Texture	Soft, crumbly, and easy to dig with your fingers
Thickness	Usually 5–20 cm deep
Contents	Humus (rotting leaves/plants), sand, silt, clay, living organisms
What lives here?	Earthworms, ants, beetles, fungi, bacteria, plant roots
Indian Example	The dark soil in your school garden or a kitchen herb pot
Why important?	All plant life — tulsi, mango, vegetables — starts here!

■ Layer 2: SUBSOIL (Middle Layer)

Colour	Light brown, yellow, or reddish — paler than topsoil
Texture	Sticky, dense, and clay-rich — harder to dig
Thickness	Usually 20–60 cm deep
Contents	Clay minerals, silt, iron/aluminum oxides, very little organic matter
What lives here?	Deep tree roots that reach down to drink stored water
Indian Example	The lighter soil visible in road cuttings in Kerala or Karnataka hills
Why important?	Stores water for big trees; provides minerals to deep-rooted plants

■ Layer 3: PARENT ROCK (Bottom Layer)

Colour	Grey, tan, or the colour of the local rock type
Texture	Hard, rough, and rocky — pebbles and broken stones
Thickness	Can extend many metres down
Contents	Broken rock, gravel, pebbles — the original material soil forms from
What lives here?	Very little — some bacteria only in cracks and crevices
Indian Example	The Deccan Plateau's basalt rock or rocky construction site diggings
Why important?	The "parent" that breaks down over thousands of years to make soil above

■ HANDS-ON ACTIVITY: Make a Core Sample!

What is a Core Sample?

A **core sample** is a cylindrical section taken from underground to study the soil layers. Scientists push a hollow tube into the ground — like pressing a straw into a layered juice drink — and pull it out with all the layers intact. We will recreate this in a bottle!

■ Materials Needed	
• 1 clear plastic bottle or glass jar (500 ml)	• Dark soil / garden soil (topsoil)
• Light sand or yellow clay (subsoil)	• Gravel, pebbles, or small stones (rock layer)
• Small leaf, twig, or toy worm (decoration)	• Stickers or paper strips + pen (for labels)
• Small spoon or scoop	• Ruler

Step-by-Step Instructions

Step 1 — Rock Layer	Add **gravel and pebbles** to the **bottom** of your bottle. Fill to about 3–4 cm. This represents the hard **Parent Rock** layer. Press it down gently. Notice how rough and hard these feel!
Step 2 — Subsoil	Carefully pour **light sand or clay** on top of the gravel layer. Fill to about 3 cm. This is the **Subsoil**. See how its colour is lighter than the soil that will go on top? That's because it has less organic matter.
Step 3 — Topsoil	Add the **dark garden soil** on top. Fill to about 3–4 cm. This is the **Topsoil**. Feel how crumbly and soft it is! This is where all the life in the soil lives.
Step 4 — Decorate	Place a small **leaf, grass blade, or toy worm** on the very top. This represents the living things that call topsoil home — earthworms, roots, and insects.
Step 5 — Label	Cut paper strips and write the layer names: **TOPSOIL** (top), **SUBSOIL** (middle), **PARENT ROCK** (bottom). Tape them to the side of the bottle at the correct heights. Your core sample is complete! ■

4. What Lives in Topsoil?

Topsoil is one of the most **biodiverse** habitats on Earth! A single teaspoon of healthy topsoil can contain more living organisms than there are people on Earth.

Organism	Role in Soil	Fun Fact
■ Earthworms	Dig tunnels, mix layers, add nutrients through droppings	Called the "farmer's best friend" — they are natural ploughs!
■ Ants	Carry food underground, aerate soil, mix layers	An ant colony can move more soil than a plough in some fields!
■ Plant Roots	Hold soil in place, absorb water & minerals	Mango roots can reach 6–8 metres deep into the soil!
■ Fungi	Break down dead material, connect plant roots	Fungal threads (mycelium) form an underground "internet"!
■ Bacteria	Convert nutrients into forms plants can absorb	Fix nitrogen from the air, fertilising soil naturally

5. Soil Science in India

Region / Soil Type	Key Feature	What Grows Here?
Deccan Plateau — Black Cotton Soil (Regur)	Rich in clay, moisture-retaining, dark coloured	Cotton, jowar, sunflower
Gangetic Plains — Alluvial Soil	Rich topsoil deposited by rivers, very fertile	Wheat, rice, sugarcane
Kerala / Karnataka Hills — Laterite Soil	Red/yellow, iron-rich subsoil visible in road cuttings	Cashew, coconut, rubber
Rajasthan Desert — Sandy Soil	Very little topsoil, large particles, drains quickly	Drought-resistant millets, cacti
Tamil Nadu Coast — Red Sandy Soil	Reddish colour from iron oxide, moderate fertility	Groundnut, mango, vegetables

■ REVIEW & QUIZ TIME!

Quick Recall

TOP layer →	TOPSOIL — dark, soft, crumbly, full of life
MIDDLE layer →	SUBSOIL — lighter, clay-rich, sticky
BOTTOM layer →	PARENT ROCK — hard stones, grey/tan
Our science tube →	CORE SAMPLE
Seeds are planted in →	TOPSOIL (where it's soft and nutrient-rich)

Quiz Questions

Q1. Which soil layer is best for growing plants, and why?

Answer:

Q2. Name two animals that live in topsoil and describe what they do.

Answer:

Q3. What colour is subsoil, and why is it different from topsoil?

Answer:

Q4. What is a core sample and how do scientists use it?

Answer:

Q5. Why do we say topsoil is like the "foundation of all plant life"?

Answer:

Q6. Name one Indian region and the type of soil found there.

Answer:

■ Take-Home Activity

Tonight, ask a family member: *"When we plant tulsi or tomatoes, which layer do the seeds go in?"* Tell them the answer is **TOPSOIL** — and explain **why** using what you learnt today! Can you point to each soil layer next time you pass a road cutting or a construction site?

Great job, Soil Scientist! ■ Remember: the ground beneath your feet is alive, layered, and full of secrets waiting to be discovered.